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Summary
A proper balance between load and recovery is crucial to initiate (optimal) physiological adaptations and avoid overuse injuries. Recovery involves physical and psychological aspects and can be enhanced through various recovery methods. The choice of a specific recovery method depends on factors like the type of activity, time until the next competition, and equipment accessibility, with considerations for the training phase and competition schedule playing a crucial role. Furthermore, different aspects of recovery follow varying individual and inter-individual time spans, making it challenging to apply group-based information to the individual level from single parameters (i.e., metabolic, neuromuscular, central, cardio-regulatory or performance-based parameters). Thus, there is a need for personalized approaches to optimize post-exercise recovery strategies. Combining group-based knowledge and repeatedly measured individual data may help practitioners to design tailored strategies.

Keywords
Regeneration – post-exercise routine – Performance – Fatigue

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Die vielfältigen Dimensionen der Erholung nach dem Sport: Die Notwendigkeit der Individualisierung

Zusammenfassung Eine ausgewogene Balance zwischen Belastung und Erholung ist entscheidend, um (otimale) physiologische Anpassungen zu initiieren und Überlastungsverletzungen zu vermeiden. Erholung bezieht sich dabei sowohl auf physische als auch psychologische As-

REVIEW / SPECIAL ISSUE

The multifaceted nature of recovery after exercise: A need for individualization

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Introduction

Training and competition loads stimulate biological adaptation, which subsequently enhance performance and potentially reduce the injury risk [45]. To ensure optimal adaptation to training and competition and to reduce the risk of overuse injuries, it is essential to find an adequate balance between load and sufficient recovery [4,45]. In this context, recovery is considered a multi-layered process, including not only a restoration of physical performance after strenuous exercise, but also affecting physiological and psychological aspects [26]. Even though the terms "regeneration" and "recovery" are sometimes used interchangeably in the literature, "recovery" is considered an umbrella term covering various aspects including regeneration and recovery strategies, while "regeneration" specifically refers to the physiological element of recovery [26]. Nevertheless, in sports, recovery after fatiguing exercise is primarily focused on physiological factors [26].

To reduce fatigue and improve recovery after exercise, athletes use various recovery techniques [27], including rest, sleep, nutrition practices (*e.g.*, administering carbohydrates, proteins, and ergogenic aids), massage, compression, stretching, hydrotherapy, and acupuncture [6]. A certain level of post-exercise fatigue is essential for functional overreaching (*i.e.*, a short-term performance decrement after exercise without signs of maladaptation and subsequently resulting in a performance enhancement). However, insufficient recovery, on the other hand, will potentially lead to non-functional overreaching [26].

To determine the best recovery strategy for an individual, the approach of 'evidence-based practice', which involves decision-

pekte und kann durch die Anwendung verschiedener Maßnahmen verbessert werden. Die Auswahl der Maßnahmen ist dabei abhängig von der Aktivität (Sportart), der Zeit bis zum nächsten Training oder Wettkampf sowie der zur Verfügung stehenden Ausrüstung. Ebenfalls bedeutsam sind die Trainings- und Wettkampfphase, die Wettkampfplanung sowie der aktuelle, individuelle Zustand des Sportlers. Darüber hinaus können verschiedene Aspekte der Erholung unterschiedliche zeitliche Verläufe sowohl innerhalb als auch zwischen Individuen aufweisen, was es schwierig macht, gruppenbasierte Informationen (z. B. metabolische, neuromuskuläre, zentrale, kardioregulatorische oder leistungsbezogene Parameter) auf individueller Ebene anzuwenden. Daher kann die Kombination von gruppenbasiertem Wissen und wiederholt gemessenen individuellen Daten Athleten und Trainern helfen, individuelle Erholungsstrategien zu entwickeln.

Schlüsselwörter
Regeneration – nach dem Training – Leistung – Ermüdung

making based on the most up-to-date evidence, may be utilized [2]. However, as different physiological systems, tissues, and performance parameters recover in different time spans [44] and may react differently (or possibly detrimental) to the application of a specific recovery method, it is essential to understand the mechanisms triggered by the application of the respective recovery methods. The impact of a recovery method varies based on factors such as the type of sport, the time of the season or training phase, and whether acute or long-term adaptations are targeted. Therefore, conventional inference-based models centered on group means may not fully reflect the complexity of the recovery process, emphasizing the need for evidence-based individualized approaches to optimize post-exercise recovery strategies.

The multifaceted nature of recovery

In general, recovery can be defined as all processes to renew an athlete's ability to regain or exceed a previous performance level, with the recovery period describing the necessary time to return to resting values after exercise [16]. The precise definition of recovery from exercises is challenging, as depending on the respective system (*i.e.*, physiological, orthopedic), recovery time may range from seconds (*e.g.*, adenosine-tri-phosphate [ATP] resynthesis through phosphocreatine breakdown) or minutes (*e.g.*, the return of acute oxygen uptake to resting levels after a bout of exercise) to weeks (*e.g.*, restoration of force-generating capacity after muscle damaging exercise) [29]. Therefore, Bishop and colleagues [4] proposed a more precise distinction of the term recovery to describe different purposes across different time spans: (i) immediate recovery (*e.g.*, ATP recovery in a leg during the swing phase between two steps), (ii) short-term recovery (*e.g.*, recovery between two strength or sprint training sets), and (iii) training recovery, describing the time between two successive work-outs or competitions [4]. In terms of recovery methods, Kellmann [25] differentiates between passive methods characterized by inactivity (e.g., rest or application of external methods such as massage), active methods aimed at metabolic processes (*e.g.*, physical activities such as cool-down running), and proactive recovery via self-determined social activities [25]. In this context, athletes judge the effectiveness of recovery methods not only based on scientific findings or the most effective technique in restoring performance [10], but also based on their perception of muscle relaxation, freshness, and reduced muscle soreness [11].

In addition to the type of activity performed, the time until the next competition or training session and the availability of equipment and staff members are relevant factors when choosing the optimal recovery method [6]. For example, in team sports, during the preseason, players mainly focus on athletic development, while during the season's competition phase, maintaining physiological condition becomes more important [23]. Maintaining the physiological

condition throughout a season is challenging, particularly when faced with a congested fixture, allowing only 3–7 days of recovery between competition and while performing additional training sessions [23]. Thus, to design an optimal recovery strategy after demanding exercises, practitioners need to consider the intensity, duration, and modality of the exercise. Furthermore, differences in training and competition regimen may require different approaches to ensure optimal regeneration.

Evidence-based application of recovery methods in the field of sports

The state of reduced physical and (or) mental performance is commonly defined as fatigue [16]. The extent of fatigue in different (physiological and psychological) systems is influenced by factors such as intensity, duration, and modality of exercise. When monitoring fatigue, often only a single parameter reflecting a specific physiological aspect is analyzed. However, as mentioned before, there may be variations in the time course of different parameters [44].

Measuring sport-specific performance is considered the gold-standard for assessing short-term fatigue, with impaired maximal strength, jump and sprint performance up to 72 h after exercise [44]. However, physiological factors and markers do not strictly follow this time course of physical performance recovery [44]. In their comprehensive review article, Skorski and colleagues [44] thus present a detailed analysis of various factors influencing the recovery process: These factors include: (I) *Metabolic factors* such as depleted phosphocreatine stores that may lead to a decreased force-generation capacity during very short recovery periods, depleted muscle glycogen stores, which may take 2–3 days being refueled after strenuous endurance exercise, or increased levels of reactive oxygen and nitrogen species that temporarily impair, but in the days following muscle damaging exercise may also act as signaling molecules to initiate muscle regeneration and adaption. Moreover, (II) *neuromuscular factors* such as task-dependent, volume- and intensity-related disruptions of the electric potential propagation, excitation-contraction coupling, and/or cross-bridging cycling. These may result in delayed onset muscle soreness and an associated lower maximal voluntary contraction or range of motion up to 96 h post-exercise (depending on the intensity, muscle action and muscle mass), with inflammation and damage markers such as creatine kinase may be being elevated above baseline levels even longer. Furthermore, (III) *central factors* such as changes in the metabolism of brain monoamines to modulate brain functions including motivation, stress responses and motor control; and (IV) *circulatory factors* such as an altered maximal excitability of the autonomic nervous system or a negative fluid balance leading to a temporary reduction in stroke volume and cardiac output, subsequently resulting in a decreased maximal oxygen uptake and thus a reduced performance.

Although the above mentioned factors are closely related to the state of fatigue and recovery, the time course of physiological markers does not necessarily follow the time course of performance recovery [44]. Therefore, when choosing a suitable recovery method to regain an athlete's performance, the intricate interplay among training load, subsequent fatigue, and adaptation plays an important role, with the chosen recovery strategy potentially influencing this dynamic positively or negatively [13]. Nevertheless, many evidence-based approaches are available to help practitioners, coaches, and athletes finding an appropriate post-exercise recovery method influencing different physiological, psychological, and performance-related parameters. Most recovery methods can be classified into nutrition-based, massage/pressure-based, or hydrotherapy-based approaches. Utilizing meta-analytical findings, Table 1 offers an overview of the primary effects of typical recovery methods.

Hydrotherapy describes different recovery techniques, including the immersion in both hot and cold water, ice massage, hot and cold showers and engagement in water-based exercises [21]. Among these approaches, cold water immersion (CWI) has gained attention for its potential to influence recovery and performance outcomes, with suggested mechanisms linked to its impact on blood flow dynamics and modulation of muscle temperature, leading to reduced post-exercise inflammation [35]. Multiple meta-analytical findings suggest that CWI is promising in mitigating the effects of muscle

Table 1. Overview of the primary effects of typical recovery methods.

Method	Outcome Parameter	Time frame	Effect size Cohen's *d* (direction)
Cryotherapy	Delayed Onset Muscle Soreness	0 h to 96 h	Trivial to Large (decrease) [22,28,35,36,48]
	Perceived Effort	0 h to 24 h	Moderate (decrease) [22,48]
	Muscular Power/Strength	0 h	Moderate to Large (decrease) [48]
		24 h to 72 h	Small to Large (increase) [28,36,48]
	Creatine Kinase	24 h to 48 h	Small to Large (decrease) [35,36,48]
	Lactate	24 h to 48 h	Moderate (decrease) [48]
Foam Rolling/Stick Massage	Delayed Onset Muscle Soreness	0 h to 96 h	Trivial (decrease) [34]
	Flexibility/Range of Motion	0 h to 96 h	Trivial to Large (increase) [34,42,46]
	Muscular Power/Strength	0 h to 96 h	Trivial (increase) [34,46]
Pressotherapy	Delayed Onset Muscle Soreness	24 h to 96 h	Small (decrease) [47]
	Muscular Power/Strength	24 h to 96 h	Trivial (increase) [47]
	Creatine Kinase	24 h to 96 h	Small (decrease) [47]
Compression Garmets	Muscular Power/Strength	0 h to 24 h	Small to Large (increase) [5]
Stretching	Delayed Onset Muscle Soreness	24 h to 72 h	Trivial (decrease) [1]
Branched Chain Amino Acids	Creatine Kinase	48 h	Moderate (decrease) [39]
	Delayed Onset Muscle Soreness	48 h	Trivial (decrease) [39]
Creatine	Creatine Kinase	48 h to 96 h	Large (decrease) [24,37]
Carbohydrate	Endurance performance	0 h	Small (increase) [33]
Protein	Muscular Power/Strength	0 h to 96 h	Small to Medium (increase) [12,38]
	Creatine Kinase	48 h to 72 h	Large (decrease) [38]
	Delayed Onset Muscle Soreness	72 h	Small (decrease) [38]

damage, as evidenced by a reduction in creatine kinase (CK) levels, alleviation of delayed onset muscle soreness (DOMS), and enhanced perceived recovery [8,21,22,28,35,36,48]. Although immediate post-exercise strength-related performance may not be positively, and in some cases, even negatively affected, an increasing body of evidence underscores the beneficial effects of CWI that become apparent within the 24h+ timeframe after resistance exercise [8,21,36]. Additionally, CWI also demonstrated positive effects on endurance performance [8,36]. To achieve optimal outcomes, water temperatures between 10 and 15 °C, coupled with immersion durations ranging from 10 to 15 min seem best [30].

However, it has been argued that the fatigue and inflammation processes following exercise might be crucial in facilitating long-term training adaptations and performance enhancements [14]. Thus, while CWI's immediate benefits are evident, the potential implications for long-term training outcomes warrant careful consideration. In this context it is noteworthy that regularly applying CWI has been associated with detrimental effects on resistance training adaptations, impacting metrics such as the one-repetition maximum, maximum isometric strength, and strength endurance performance, while the influence on aerobic exercise performance appears less pronounced, with no discernible impact on time-trial mean power and distance or maximal aerobic power during graded exercise tests [31]. Nevertheless, at least for whole body cryotherapy (WBC) induced by cold chambers, there also seems to be a positive effect for a chronic application: after a sufficient number (10+) of WBC sessions the stimulation of immunological responses has been reported to increase the

body's immunity with a simultaneous decrease in total oxidative status and inflammatory responses [15].

Although compression has been shown to significantly reduce muscle soreness [47], foam rolling, stick massage, and pressotherapy have no significant effect on range of motion, muscle swelling, and recovery of muscle strength, jumping performance, and maximal voluntary isometric contraction up to 96 h after muscle-damaging exercise [34,47]. However, foam rolling shows a (statistically non-significant) trivial tendency towards better strength recovery [46]. On the other hand, compression garments seem to positively impact strength recovery within the first 24 h after exercise [5].

In terms of nutritional aspects, at least in endurance training, the consumption of carbohydrates has been shown to have a positive impact on performance immediately following the training session (up to 4 h) and even in the days following the session [9,33]. Interestingly, the additional ingestion of proteins during these time frames does not seem to offer any additional performance benefits [9,33]. Regarding resistance training, meta-analytical findings suggests that both creatine and branched-chain amino acids (BCAAs) intake reduce creatine kinase levels for up to 96 h after exercise [24,38,39]. However, in terms of effects on other aspects, such as muscle soreness (BCAA, creatine, protein intake), muscle strength, range of motion, and inflammation (creatine), the evidence is inconclusive [37,39]. Nevertheless, in the context of resistance training, it's worth noting that the consumption of protein peri-exercise may play a role in preserving both isometric and isokinetic maximal voluntary contraction in the 96 h post-exercise [38]. This phenomenon could be attributed to the observed small to medium effects of protein intake in aiding the restoration of contractile function within this specific time frame [12]. Moreover, considering the long-term effects of resistance training over 6+ weeks, protein supplementation positively influences the increase of fat-free mass and one-repetition-maximum strength [7].

Overall, different recovery methods distinctly affect specific factors of recovery over different time courses [44]. For example, in a recent inverstigation, participants performed exhaustive eccentric exercise (*i.e.*, depth jumps from a height of 50 cm) to initiate muscular fatigue. Immediately after the exercise protocol, participants were randomly allocated to receive different recovery methods (Cold Water Immersion, Percussive Massage or Passive Rest). To quantify the recovery state of the subjects, a variety of physiological (*e.g.*, create kinase), performance-related (*e.g.*, jumping performance) and psychological metrics (*e.g.*, perceived muscle stiffness) were obtained immediately before and after the fatiguing protocol, as well as immediately, 24 h, 48 h and 72 h after the recovery protocol. Even tough on a group-based level large and significant method × time-interaction effects have been found with large differences especially immediately after the recovery protocol, individual time course of recovery may differ distinctively (Fig. 1).

Evidence-based individualization of recovery methods

In the scientific literature, the effect of different recovery methods has often been investigated by employing randomized controlled designs (RCT) with the focus on the main effects of the (sample) mean differences in pre-post changes [18] in metabolic, neuromuscular, central, cardio-regulatory or performance-based parameters [44]. However, the problem that emerges when applying group-based results to the individual is heavily discussed in medicine: For example, a drug being considered marginally effective in the general population is not necessarily similarly effective in any given subpopulation [40,41]. This problem occurs in a similar form when evidence-based decisions are to be made for the use of training and/or recovery techniques in elite sports. Considering the small number of elite athletes and the possibly high relevance of even small effects, there are only two suboptimal solutions when designing RCTs in elite sports: (i) justifying the small sample size by mentioning the elite status of the included participants and thus, due to low statistical power, possibly failing to detect an effect or find a hard to replicate positive result by overestimating the effect size, or (ii) reducing the required performance level of athletes to be eligible to the study, subsequently hardening the trans-

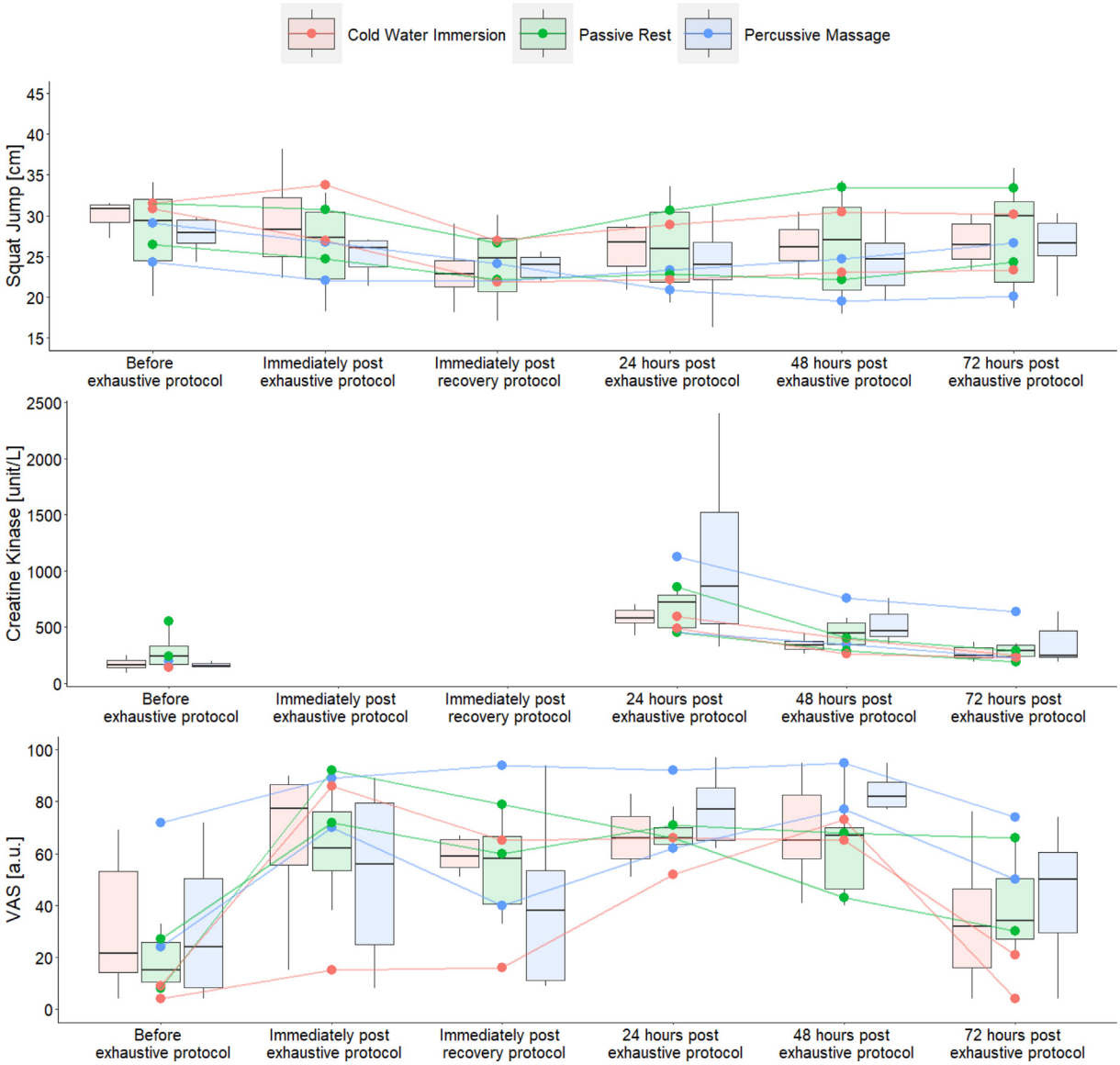

Figure 1
Boxplots of (top to bottom) Squat Jump height, blood creatine kinase levels and perceived leg stiffness rated on a visual analogue scale (VAS) for the Cold Water Immersion group (n = 11, blue), Percussive Massage group (n = 11, red) and Passive Rest group (n = 10, green) at the different time points. Furthermore, individual time courses of the different factors for two exemplary individuals of each group deviating substantially from the group mean are also indicated (scatter + lines).

ferability of the findings to the elite level [43].

However, even when conducting an adequately powered RCT with a justifiable selection of homogenous participants to estimate the effect of a specific treatment, the "true effect" of a treatment differs from the observed as the data is affected by different types of error [18]: the random error (*i.e.*, the background noise causing unexplained variation between repeated measures), the between-subject-variability (*i.e.*, the difference in values between subjects) and the within-subject-variability (*i.e.*, the variability in pre-post differences

without the random error) [18]. Furthermore, just as the adaptation to a standardized training is affected by many factors including nutrient accessibility, age, hereditary and phenotype-linked elements [32], it is reasonable to assume that variations in recovery outcomes will also manifest themselves among individuals. To identify the individual consistent response of an athlete to a specific training or recovery technique ("subject-by-training interaction"), the study designs of repeated testing or repeated interventions with a statistical approach of linear models may be incorporated [18]. For this, Hecksteden and colleagues [18] propose two different approaches: The first approach postulates the repetition of an intervention with adequate washout periods. In contrast, the second approach involves repeated testing of the respective outcome measure during the intervention. Thus, via repeated measurements of blood markers, individualized reference ranges for monitoring muscle recovery have been created [20] and implemented in a real-life setting to closely monitor individual muscle recovery in elite badminton players [3].

Furthermore, by including pre-existing knowledge generated from group-based information as an informative prior into a Bayesian analysis, the effectiveness of specific recovery methods can be tested for individuals via the repeated application of said method [17]. This combination of group-based information amended by individual information allows for a more precise characterization of an individual athlete [19], subsequently helping to decide on the best available recovery strategy.

Conclusion

In summary, when designing effective recovery strategies, it is important to define short- and long-term objectives, accounting for the intricate physiological, metabolic, and cardio-circulatory implications of exercise while evaluating the potential impact of various recovery approaches. Although group-based effects highlighted in randomized controlled trials offer insights within (sub-elite) populations, their translation to individual cases remains uncertain. Thus, the selection of appropriate recovery strategies hinges upon evidence-based methods with well-documented effects subjected to personalize testing through repeated applications and vigilant monitoring. Given the diverse desired outcomes, a discerning approach is vital, employing distinct recovery techniques tailored to specific contexts, such as competitive scenarios or training phases. Ultimately, choosing effective recovery strategies requires a holistic understanding that includes immediate and enduring goals, complex physiological considerations, and closely monitored application of recovery methods, while taking into account the variable nature of individual responses.

CRediT authorship contribution statement

Ludwig Rappelt: Conceptualization, Investigation, Writing – original draft, Visualization. **Sasha Javanmardi:** Conceptualization, Investigation, Writing – review & editing. **Lars Heinke:** Investigation, Writing – review & editing. **Christian Baumgart:** Conceptualization, Visualization, Writing – review & editing, Supervision. **Jürgen Freiwald:** Conceptualization, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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